

Activation of fast skeletal muscle troponin as a potential therapeutic approach for treating neuromuscular diseases

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Limited neural input results in muscle weakness in neuromuscular disease because of a reduction in the density of muscle innervation, the rate of neuromuscular junction activation or the efficiency of synaptic transmission. We developed a small-molecule fast-skeletal-troponin activator, CK-2017357, as a means to increase muscle strength by amplifying the response of muscle when neural input is otherwise diminished secondary to neuromuscular disease. Binding selectively to the fast-skeletal-troponin complex, CK-2017357 slows the rate of calcium release from troponin C and sensitizes muscle to calcium. As a consequence, the force-calcium relationship of muscle fibers shifts leftwards, as does the force-frequency relationship of a nerve-muscle pair, so that CK-2017357 increases the production of muscle force in situ at sub-maximal nerve stimulation rates. Notably, we show that sensitization of the fast-skeletal-troponin complex to calcium improves muscle force and grip strength immediately after administration of single doses of CK-2017357 in a model of the neuromuscular disease myasthenia gravis. Troponin activation may provide a new therapeutic approach to improve physical activity in diseases where neuromuscular function is compromised.